

Brass City, Burdened Water: The Industrial Water-Pollution Geography of Moradabad and a Framework for Governing It

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Abstract:

Moradabad, in western Uttar Pradesh, is known worldwide as the Brass City, and its brassware, electroplating, and metal-recycling clusters sustain the regional economy. The same industry discharges metal-laden effluent that has left a documented mark on the district's water: heavy metals including arsenic, chromium, copper, nickel, lead, and zinc accumulate in the sediments of the Ramganga-Ganga interfluvium and threaten the shallow groundwater on which most of the district depends. This paper offers a geographical treatment of that pollution and a framework for governing it. It argues that Moradabad's contamination is best understood spatially, as a source-pathway-receptor system with a distinctive and dangerous feature: the closed interfluvium basins that act as sinks, concentrating metals over time and delivering them slowly toward the aquifer rather than flushing them downstream. Read this way, the problem is not a diffuse nuisance but a set of identifiable sources, pathways, and exposed populations whose spatial relationships determine where harm occurs and where intervention will work. The paper proposes a governance framework built on cluster-scale effluent treatment, spatial zoning that separates polluting activity from recharge-sensitive ground, source-to-receptor monitoring, and an approach to the large informal component of the industry that regulation alone cannot reach. The framework is illustrated for Moradabad and is transferable to the many artisanal-industrial towns of the Global South where a livelihood-giving industry and the water it contaminates occupy the same ground.

Keywords: industrial water pollution; heavy metals; brass industry; Ramganga; environmental governance; Moradabad

1. Introduction

Few Indian cities are as closely identified with a single industry as Moradabad is with brass. Known as Peetal Nagri, the Brass City, it has for generations produced the brassware and metal handicrafts that support much of the regional economy and reach markets across the world. That industry, together with the electroplating and metal-recycling activities clustered around it, is also a source of water pollution, discharging effluent rich in the heavy metals used and released in metalworking. The result is a documented contamination of the district's water environment: analyses of the sediments of the Moradabad area, in the interfluvium of the Ganga and Ramganga rivers, have found significant concentrations of arsenic, chromium, copper, nickel, lead, and zinc, with the pollution concentrated where industrial and urban inputs are highest (Sharma et al., 2003). Because the great majority of the district's population relies on groundwater, and because that groundwater is hydraulically connected to the contaminated surface environment, the pollution is not only an ecological problem but a public-health one.

This paper treats Moradabad's industrial water pollution as a geographical problem and proposes a framework for governing it. The geographical framing matters because contamination is not spatially uniform: it originates at specific industrial sources, travels along specific pathways of drains, rivers, and sediments, and reaches specific exposed populations, and the spatial relationships among these determine both where harm concentrates and where intervention will be effective. A distinctive and troubling feature of the Moradabad setting sharpens this point. The vast interfluvium contains numerous closed basins that rarely connect to any river and so act as sinks, within which sediments and their metal load are redeposited rather than flushed away, so that concentrations rise over time and threaten to migrate laterally and vertically into the aquifer (Sharma et al., 2003). Pollution here is thus not carried off downstream but banked in place, a dynamic that any governance response must confront.

The paper first sets out the pollution problem as a source-pathway-receptor system, then develops a governance framework of four components, and then illustrates both for Moradabad. The contribution is conceptual and synthetic: it organises documented evidence into a spatial framework and derives governance implications from it, rather than reporting new measurements. The aim is to show how a geographer's spatial reasoning can turn a known contamination problem into a governable one, in Moradabad and in the many towns like it.

2. The pollution problem

2.1 The source: an industry of metals

The origin of the problem is the metal-based industry itself. Brassware manufacture, electroplating, and the recycling of metal and electronic waste all use and release heavy metals, and where their effluent is discharged without adequate treatment it carries those metals into the water environment. In Moradabad this industry is not a single large emitter but a dense fabric of enterprises, many of them small and informally organised, distributed through the urban area and its surroundings. This structure is central to the problem: a pollution load that arises from many small, dispersed, and often unregistered sources is far harder to regulate than one from a few large plants, and it resists

the conventional model of point-source control. The source, in other words, is not only metallic but organisational, and its dispersed and partly informal character shapes everything downstream.

2.2 The pathways: drains, river, and the closed-basin sink

From these sources the metals travel along definable pathways. Untreated effluent enters drains and reaches the Ramganga, whose pollution load is highest where industrial and urban inputs concentrate, though the flowing river also dilutes and disperses its burden downstream (Sharma et al., 2003). The more insidious pathway is the one that does not flow. Across the interfluvium, closed basins receive and retain sediment, and because they neither connect to a river nor flush their contents, they accumulate metals within themselves, acting as sinks in which concentrations increase over time. From these sinks the metals threaten to move slowly into the shallow aquifer beneath. This distinction between the flushing pathway of the river and the accumulating pathway of the closed basin is the analytical heart of the problem: contamination that would be carried away in a well-drained landscape is instead stored and concentrated here, turning the interfluvium into a reservoir of pollution poised above the groundwater.

2.3 The receptor: a groundwater-dependent population

The pathways matter because of who lies at their end. The population of the district depends overwhelmingly on groundwater for domestic use, and to a significant degree for agriculture, so the aquifer is the receptor whose contamination translates directly into human exposure. This dependence on groundwater is characteristic of the wider Indo-Gangetic plain, where the resource is both the foundation of water security and, increasingly, a vector of contamination (MacDonald et al., 2016). Where the closed-basin sinks deliver their metal load into that aquifer, and where irrigation draws contaminated water onto food crops, the exposure pathway to the population is complete. The receptor, in the geographical framing, is not an abstraction but the specific groundwater-dependent communities whose wells and fields lie within reach of the contaminated pathways.

3. Approach

The paper synthesises documented evidence into a spatial framework rather than generating new data. Its method is to organise Moradabad's pollution as a source-pathway-receptor system, the standard structure for reasoning about environmental contamination, and to give that structure explicit geographical content: where the sources are and how they are organised, along which pathways the contaminants move and how those pathways differ in their behaviour, and which populations occupy the receptor end. From this spatialised understanding the governance framework is then derived, on the principle that effective intervention must act at identifiable points in the source-pathway-receptor chain rather than on pollution in general. The framing draws on the water-security perspective, which treats acceptable water quality and acceptable water-related risk as integral to security (Grey & Sadoff, 2007), and on the nexus perspective, which insists that a problem joining industry, water, health, and livelihood be governed across those domains together rather than sectorally (Hoff, 2011).

4. A spatial framework for the pollution

Organising the evidence as a source-pathway-receptor system yields a framework with clear governance leverage, because each element of the chain is a place where the flow of harm can be interrupted. At the source, the leverage lies in the effluent of the metal industry, and the governance challenge is the dispersed and partly informal organisation of that industry. Along the pathways, the leverage differs sharply between the flowing river, where dilution offers some natural mitigation but downstream communities inherit the load, and the closed basins, where no such mitigation occurs and the accumulating store must be treated as a growing hazard in its own right (Sharma et al., 2003). At the receptor, the leverage lies in protecting the aquifer and the exposed population, through the safeguarding of recharge-sensitive ground and the monitoring of drinking and irrigation water. The framework's central insight is that these leverage points are spatially specific: the closed-basin sinks in particular are identifiable locations where contamination is concentrating and from which it threatens the groundwater, and they demand a targeted response that a spatially blind approach to pollution would never identify. Seeing the pollution as a spatial system is therefore not a description but a guide to action.

5. A governance framework

5.1 Cluster-scale effluent treatment

The first component addresses the source. Because the pollution arises from many small enterprises rather than a few large ones, individual treatment at each unit is impractical, and the appropriate response is treatment organised at the scale of the industrial cluster, shared infrastructure that receives and treats the combined effluent of many small producers. This aligns the scale of the solution with the scale of the source and brings within reach of treatment the dispersed emitters that unit-level regulation cannot practically cover. The design of such shared treatment, its capacity, siting, and the arrangements for connecting informal producers to it, is the crux, and it is where the organisational character of the industry must be met directly.

5.2 Spatial zoning of activity and recharge

The second component addresses the pathways, and particularly the closed-basin danger. If contamination concentrates where polluting activity sits above or drains toward recharge-sensitive ground, then separating the two in space is a direct governance lever. Zoning that keeps metal-based industry and its effluent away from the closed basins and other recharge-sensitive areas, and that treats those areas as protection priorities, attacks the pathway at the point where it turns toward the aquifer. This is the spatial-planning counterpart to effluent treatment: the one reduces the load at the source, the other manages where that load can and cannot go, and together they address both the quantity of contamination and its geography.

5.3 Source-to-receptor monitoring

The third component makes the system visible and accountable. Monitoring designed along the source-pathway-receptor chain, at the points of discharge, in the river and the closed basins, and in the groundwater and irrigation water at the receptor end, would track where contamination originates, how it moves, and whether it is reaching people, and would provide the evidence base

that targeted governance requires. Without such monitoring, intervention is guesswork and accountability is impossible; with it, the closed-basin sinks can be watched as the hazards they are, and the effect of treatment and zoning can be measured. Monitoring is thus not an adjunct to the framework but the instrument that makes its other components operable and answerable.

5.4 Engaging the informal sector

The fourth component confronts the reality that a large part of Moradabad's metal industry operates informally, beyond the effective reach of conventional regulation. A governance approach that relies on enforcement against registered units alone will miss much of the source, so the framework must include means of engaging the informal sector: connecting small and unregistered producers to shared treatment, providing the support and incentives that make cleaner practice feasible for enterprises operating at the margins, and recognising that the same industry which pollutes the water also sustains the livelihoods of many who depend on it. This is where the nexus character of the problem is sharpest, since water, industry, health, and livelihood meet in the same small workshops, and governance must hold those objectives together rather than sacrificing one to another (Hoff, 2011; Grey & Sadoff, 2007).

6. Illustrative application: Moradabad

The framework can be read through Moradabad itself, where its elements are already present. The source is the city's dispersed and partly informal brass, electroplating, and metal-recycling industry; the pathways are the drains and the Ramganga on one hand and the accumulating closed basins of the interfluvium on the other; and the receptor is the district's groundwater-dependent population (Sharma et al., 2003; MacDonald et al., 2016). Applying the framework, the closed basins would be identified and treated as priority hazards, monitored for their metal load and their threat to the aquifer beneath; cluster-scale effluent treatment would be developed to bring the many small producers within reach of treatment; zoning would separate polluting activity from the recharge-sensitive ground above the aquifer; source-to-receptor monitoring would track contamination from workshop to well; and engagement with the informal sector would seek to reconcile the protection of water with the livelihoods the industry provides. The result would be a response matched to the actual geography of the problem rather than to pollution in the abstract. Consistent with the paper's conceptual aim, this illustrates how the framework would operate rather than reporting new measurements, which implementation would require.

7. Discussion

The framework's claim is that Moradabad's industrial water pollution becomes governable when it is understood spatially, as a source-pathway-receptor system whose most dangerous feature is the closed-basin sink that banks contamination in place above the aquifer. Three implications follow. First, the spatial framing identifies leverage that a non-spatial view misses, above all the closed basins as targetable hazards rather than diffuse background pollution (Sharma et al., 2003). Second, the governance response must match the organisational geography of the source, which means cluster-scale rather than unit-scale treatment and the direct engagement of an informal sector that regulation alone cannot reach. Third, the problem is inescapably a nexus one, joining a livelihood-

giving industry, a contaminated water resource, and an exposed population, so that governance must protect water and health without destroying the economic base of the community, holding those aims together (Hoff, 2011; Grey & Sadoff, 2007; MacDonald et al., 2016). The framework is transferable: artisanal and small-scale industrial clusters that pollute the water on which their own workers depend are common across the Global South, and the source-pathway-receptor logic, with attention to local sink dynamics, applies wherever they occur. Limitations are clear. The framework is conceptual and rests on existing evidence; establishing the current extent of contamination and the behaviour of specific basins would require the monitoring the framework itself calls for, and the political economy of regulating an informal, livelihood-critical industry is formidable. These define the agenda for empirical and policy work, and they are the reason the framework treats monitoring and informal-sector engagement as central rather than peripheral.

8. Conclusion

Moradabad's brass industry gives the city its identity and much of its livelihood, and it burdens the city's water with the metals of its trade. This paper has argued that the resulting pollution is best understood and governed spatially, as a source-pathway-receptor system in which the closed basins of the Ramganga-Ganga interfluvium act as sinks that concentrate contamination and threaten the groundwater on which the district depends. The governance framework it proposes, cluster-scale effluent treatment, spatial zoning of activity and recharge, source-to-receptor monitoring, and engagement with the informal sector, is matched to that geography and to the organisational reality of the industry. Illustrated for Moradabad and transferable to artisanal-industrial towns across the Global South, the framework offers a way to protect water and health without abandoning the livelihoods the industry sustains. Its promise rests on implementation, beginning with the monitoring that would make the geography of the problem fully visible.

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